The Great Space Race
Problem ID: thegreatspacerace

You are participating in the biggest race in the galaxy and you are certain your newly developed warp drive will be able to pull home the win. However, the fuel cost per lightyear traveled is astronomical, and you are on a budget. Since you would prefer to keep as much of prize as possible, you need to figure out how to minimize the fuel cost.

This is where there are both good and bad news. Even if you totalled up the cost of all fuel canisters you could possibly need, they are still cheaper than even traveling one light year too far, and you have enough capacity to carry however many will be needed. However, the fuel canisters only come in certain specific sizes, though there is luckily a limitless supply you can buy from.

As such, you need to figure out what combination of fuel canisters will get you as close to the finish line as possible while at least reaching it. And since you still want to get as much of the prize as possible, you want to minimize the amount of fuel canisters needed as well.

Input

The first line of the input consist of an integer $0 <= n <= 10^5$ indicating how much fuel is needed to reach the goal. The next line of input consist of i number of space seperated integers indicating the different sizes of fuel canisters. Each fuel canister has a size of $0 <= s <= 10^5$. If $n > 0$ and there is only 1 fuel canister then the fuel canister is guarenteed to be of a size larger than 0.

Output

Each line of the ouput should consist of two space seperated integers: the number of fuel canisters of a specific size followed by the specific fuel canister size. Every fuel canister size from the input should appear in the ouput. If there are multiple valid solutions with the same number of canisters used, any one of them will be accepted.

Sample Input 1
```
10
1 2 3
```

Sample Output 1
```
3 3
1 1
0 2
```

Sample Input 2
```
500
0 101 102 103
```

Sample Output 2
```
101 5
102 0
103 0
```

Sample Input 3
```
900
23 74 129 2 1 53 88
```

Sample Output 3
```
0 23
4 74
4 129
0 2
0 1
0 53
1 88
```